

The influence of early moderate prenatal alcohol exposure and maternal diet on offspring DNA methylation: a cross-species study

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2
March 26, 2024 (this version)

Reviewed preprint version 1
December 7, 2023

Sent for peer review
September 26, 2023

Posted to preprint server
August 24, 2023

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Abstract

Alcohol consumption in pregnancy can affect genome regulation in the developing offspring but results have been contradictory. We employed a physiologically relevant murine model of short-term moderate prenatal alcohol exposure (PAE) resembling common patterns of alcohol consumption in pregnancy in humans. Early moderate PAE was sufficient to affect site-specific DNA methylation in new-born pups without altering behavioural outcomes in adult littermates. Whole-genome bisulfite sequencing of neonatal brain and liver revealed stochastic influence on DNA methylation that was mostly tissue-specific, with some perturbations likely originating as early as gastrulation. DNA methylation differences were enriched in non-coding genomic regions with regulatory potential indicative of broad effects of alcohol on genome regulation. Replication studies in human cohorts with fetal alcohol spectrum disorder suggested some effects were metastable at genes linked to disease-relevant traits including facial morphology, intelligence, educational attainment, autism, and

schizophrenia. In our murine model, a maternal diet high in folate and choline protected against some of the damaging effects of early moderate PAE on DNA methylation. Our studies demonstrate that early moderate exposure is sufficient to affect fetal genome regulation even in the absence of overt phenotypic changes and highlight a role for preventative maternal dietary interventions.

eLife assessment

This **important** study unveils the significant impact of prenatal alcohol exposure on epigenetic patterns, offering new insights into its adverse health outcomes through **solid** evidence from both mouse models and human data. The findings, which reveal how a high-methyl diet can mitigate these epigenetic alterations, present a promising prenatal care strategy. Despite its **solid** data overall, the study's small sample size and unaccounted confounders suggest the need for further research to confirm these findings and explore their practical implications.

Introduction

Alcohol consumption in pregnancy is a leading cause of neurodevelopmental impairments in children (1). Alcohol can pass through the placenta acting as a teratogen in fetal tissues causing physical, cognitive, behavioural, and neurodevelopmental impairment in children at high doses with lifelong consequences for health (2). Fetal Alcohol Spectrum Disorder (FASD) and Fetal Alcohol Syndrome (FAS) can arise at binge levels of exposure, although not always at lower levels of exposure. Whether PAE is sufficient to induce overt physiological abnormalities depends on multiple environmental and genetic factors including the dose and timing of alcohol use during pregnancy, maternal diet, smoking, stress, and potentially other factors that collectively influence fetal outcomes (2–4).

Patterns of alcohol consumption in pregnancy vary, but epidemiological surveys suggest most women in Western countries consume low to moderate levels between conception until recognition of pregnancy (5), after which time consumption largely ceases, apart from occasional use (6). While the effects of binge levels of exposure are well documented as able to cause FASD, more subtle effects that reflect the more common patterns of drinking are unclear and more research is needed to support public health initiatives to reduce alcohol consumption in pregnancy.

Studies suggest alcohol can disrupt fetal gene regulation through epigenetic mechanisms (7). DNA methylation is one epigenetic mechanism involving the catalytic addition of methyl groups to cytosines bases within cytosine-guanine (CpG) dinucleotide motifs during one-carbon metabolism. Methylation of DNA can alter chromatin density and influence patterns of gene expression in a tissue-specific and developmentally appropriate manner and disruption to this process may cause some of the difficulties experienced by people with FASD (8, 9). Previous studies on human participants (10–13) and animals (14–16) report that alcohol can disrupt DNA methylation either globally (12–15), and/or at specific gene regions (11, 15, 16). Our recent systematic review, however, found limited replication of effects between studies suggesting the effects of alcohol on DNA methylation may be stochastic and influenced by numerous confounding factors (7). PAE can either directly inhibit DNA methyltransferase enzymes or disrupt one-carbon metabolism via inhibition of bioavailability of dietary methyl donors, such as folate and choline to the fetus (17, 18). Choline, in particular, has been explored in several clinical trials to reduce cognitive deficits caused by PAE in affected individuals (19, 20), or

when administered during pregnancy (21 [↗](#), 22 [↗](#)), with results suggesting a high methyl donor diet (HMD) could at least partially mitigate the adverse effects of PAE on various behavioural outcomes.

Given the lack of clarity around the effects of typical patterns of alcohol consumption, which often do not cause observable phenotypes, we conducted an epigenome-wide association study of early moderate PAE in mice. Regions identified as sensitive to gestational alcohol exposure were replicated in human cohorts. The study was a controlled intervention investigating the impact of early moderate PAE on offspring DNA methylation comparing exposed and unexposed mice, with an additional arm comparing the effect of alcohol exposure in the context of a high methyl donor maternal diet. The exposure period covers the equivalent of pre-conception up until the first trimester in humans when neurulation occurs, reflecting a typical situation in which women may consume alcohol up until pregnancy recognition (5 [↗](#))(23 [↗](#)). The primary outcome of the study was differences in offspring DNA methylation and secondary outcomes of behavioural deficits across neurodevelopmental domains relevant to FASD were also examined. We employed whole-genome bisulfite sequencing (WGBS) for unbiased assessment of CpG DNA methylation in newborn brain and liver, two target organs affected by ethanol (24 [↗](#)), to explore tissue-specificity of effects and to determine any ‘tissue agnostic’ effects which may have arisen prior to the germ-layers separating in early gastrulation. We also conducted candidate gene testing of regions identified in prior studies as sensitive to early moderate PAE. Our study provides cogent evidence that common patterns of drinking can have measurable effects on fetal gene regulation, highlighting a role for maternal dietary support in public health interventions.

Results

Comparison of prenatal characteristics across treatment groups

To investigate the effects of early moderate PAE and a high-methyl donor diet (HMD) across pregnancy on offspring DNA methylation and behavioural outcomes, we employed a murine model with four treatment groups. Treatment groups were designed to assess the effect of alcohol on offspring DNA methylation compared to control mice. An additional treatment group included a maternal diet high in methyl donors, with and without alcohol exposure (**Figure 1** [↗](#)).

The trajectory of weight gain during pregnancy was similar across all treatment groups with some evidence of more rapid weight gain in the HMD groups in the last 2 days (Linear mixed effects regression model; H₂O-HMD: -2.282 ± 0.918 g, $P = 0.0177$; PAE-HMD: -1.656 ± 0.814 g, $P = 0.0493$; **figure S1a** [↗](#)), although the total amount of weight gained between days 1 and 17-19 was not significantly different between treatment groups by linear mixed effects regression (**Figures S1a-b** [↗](#)). The total amount of liquid consumed over the course of pregnancy was significantly lower in HMD dams by unpaired t-test (**Figures S1b** [↗](#) and **S2c** [↗](#)). There was no significant difference in the average litter size (6.525 ± 0.297 pups, **figure S1c** [↗](#)) and pup sex ratios (**Figure S1d** [↗](#)) between treatment groups by unpaired t-test.

Effects of early moderate PAE on DNA methylation in newborn mice tissue

A subset of 16 newborn pups ($n = 4$ per treatment) matched for sex and litter size were selected for whole-genome bisulfite sequencing. Both neonatal brain and liver were harvested to investigate the effects of early moderate PAE on fetal CpG DNA methylation. A total of 21,842,961 CpG sites were initially available for analysis.

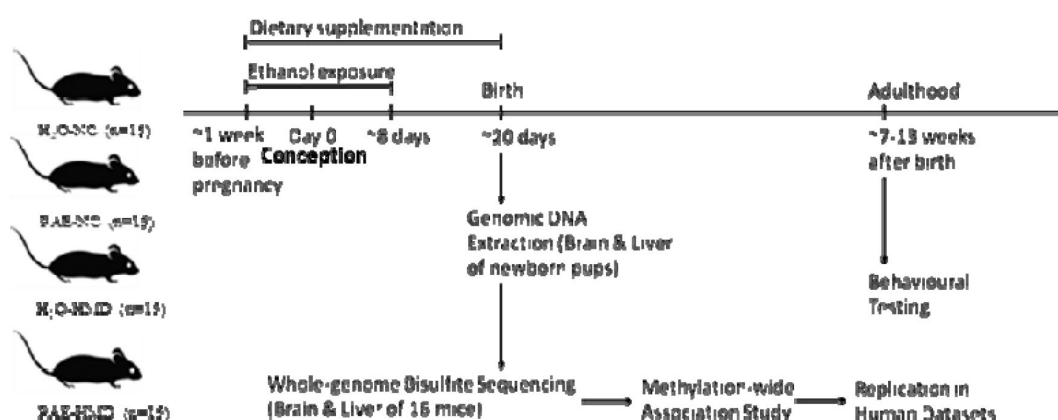


Figure 1.

Overview of PAE model.

A schematic representation of the experiment design is shown in [figure 1](#). Fifteen dams were allocated to each treatment group. Prenatal alcohol exposure (PAE) mice were exposed to ethanol (10% v/v in non-acidified reverse osmosis drinking water *ad libitum*) from one week before pregnancy to gestational days 8-10 and the remaining mice received water (H₂O). The PAE and H₂O groups received either normal chow (NC) or a high methyl donor (HMD) diet (NC containing 20 mg/kg folate and 4970 mg/kg choline) from one week before pregnancy until birth.

Global levels of DNA methylation stratified across different genomic contexts were preserved across treatment conditions, with no major differences in average DNA methylation content between groups (**Figure S4**). To investigate region-specific effects of early moderate PAE on newborn DNA methylation, we conducted genome-wide testing comparing exposed and unexposed mice on the normal chow. We identified 78 differentially methylated regions (DMRs) in the brain and 759 DMRs in the liver ($P < 0.05$ and mean difference in methylation across the DMR with PAE (Δ) > 0.05) from ~19,000,000 CpG sites tested after coverage filtering (**Figure 2a-b**). These regions were annotated to nearby genes using *annotatr* and are provided in Tables S1-2. Two of the DMRs overlapped in mouse brain and liver (tissue agnostic), but the remainder were tissue specific. Among these tissue agnostic regions was the *Impact* gene on chromosome 18, which had lower methylation in PAE+NC mice compared to H₂O+NC mice in both the brain and liver (**Figure 2c-d**). The other tissue agnostic region was within 5kb downstream of *Bmf* and had higher DNA methylation in brain and liver tissue of PAE+NC mice.

Lower DNA methylation with early moderate PAE in NC mice was more frequently observed in liver DMRs (93.5% of liver DMRs), while brain DMRs were almost equally divided between lower and higher DNA methylation with early moderate PAE (52.6% of brain DMRs had lower DNA methylation with early moderate PAE). Some DMRs localised to the same genes in both brain and liver, although they were different regions. The three genes affected by PAE in both brain and the liver tissues were the Autism Susceptibility Gene 2 (*Auts2*), Androglobin (*Adgb*), and RNA Binding Protein Fox 1 (*Rbfox1*) genes (**Table 1**). In both brain and liver tissues, DMRs were enriched in non-coding intergenic and open sea regions and relatively underrepresented in coding and CpG island regions (**Figure 2e-f**). Using open chromatin assay and histone modification datasets from the ENCODE project, we found enrichment ($p < 0.05$) of DMRs in open chromatin regions (ATAC-seq), enhancer regions (H3K4me1), and active gene promoter regions (H3K27ac), in mouse fetal forebrain tissue and fetal liver (**Table 2**). Gene ontology enrichment analysis of liver DMRs that did localise to genes showed enrichment in ten predominantly neuronal pathways, with neuron projection being the most significant (**Figure 2g**, Tables S3-4).

HMD mitigates the effects of early moderate PAE on DNA methylation

To determine whether administration of a HMD throughout pregnancy could mitigate the effects of PAE on offspring DNA methylation, we examined alcohol sensitive DMRs identified in the previous analysis in the HMD mice. Compared to control mice (H₂O+NC), PAE+HMD mice exhibited significant ($p < 0.05$) DNA methylation differences in only 12/78 (15%) brain (Table S7), and 124/759 (16%) liver (Table S8) DMRs, suggesting the effects were predominantly mitigated. Effect sizes compared to mice on the normal chow were substantially lower, in some cases more than 25% reduced in mice on the high methyl donor diet (**Figure 3**).

Effects of early moderate PAE and HMD on behavioural outcomes in adult mice

Remaining littermates from each treatment group were reared to adulthood and underwent behavioural testing assessing various neurocognitive domains that can be affected in FASD including locomotor activity, anxiety, spatial recognition, memory, motor coordination and balance. There was no evidence that early moderate PAE had a significant effect on any of the behavioural outcomes tested (**Figure S3**). Mice exposed to HMD exhibited greater locomotor activity, in terms of distance travelled (**Figure 4**).

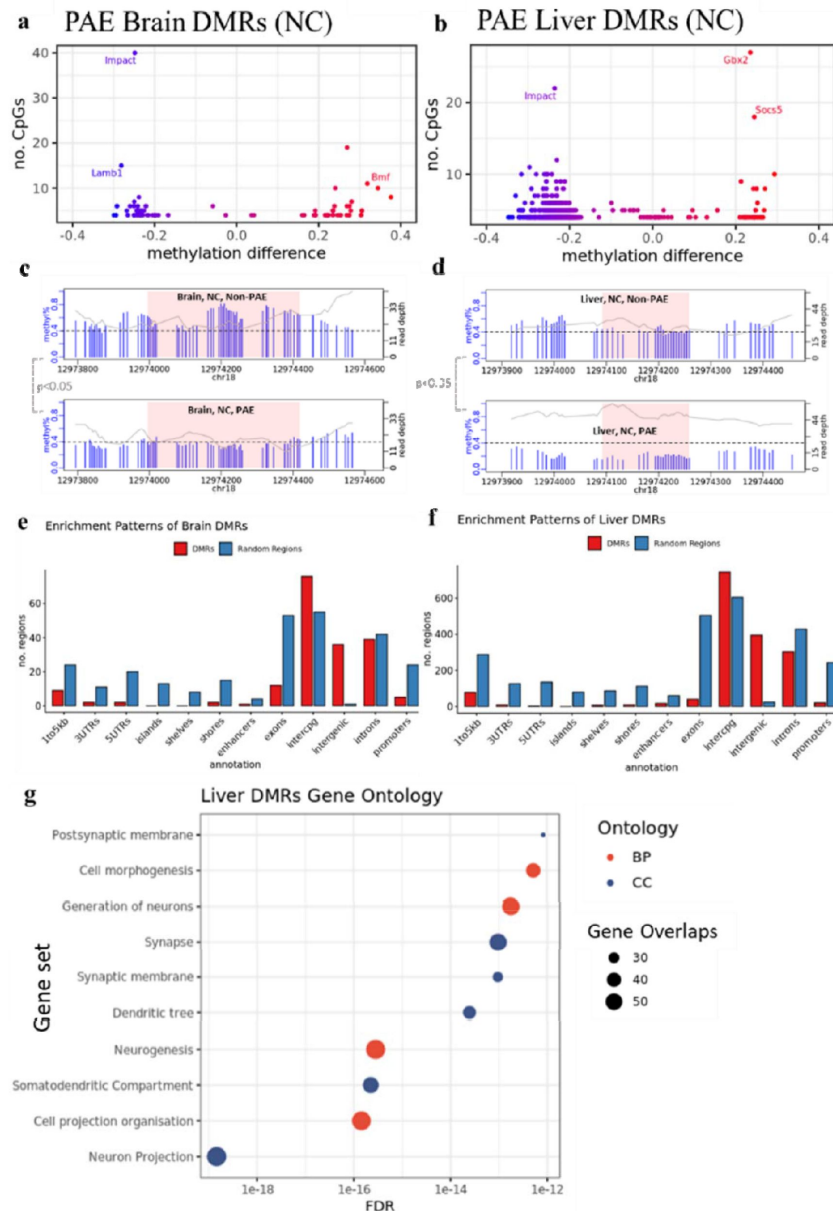


Figure 2.

PAE was associated with site-specific differences in offspring DNA methylation.

The majority of DMRs lost methylation with PAE in (a) brain and (b) liver of mice given normal chow. Each point represents one DMR. Point colour indicates change in DNA methylation with PAE. PAE was also associated with lower methylation in the DMRs identified in the promoter of the *Impact* gene in (c) brain and (d) liver, within NC mice. Each plot represents a separate treatment group. Each blue vertical line indicates a CpG site, with the height and corresponding left y-axis indicating the methylation ratio. The grey line and corresponding right y-axis indicate coverage at each CpG site. The black horizontal dotted line indicates 40% methylation for comparison purposes. The x-axis indicates the base position on chromosome 18, with the pink shaded area highlighting the DMR. DMR plots include 200 base pair flanking regions on each side of the DMR. DMRs identified in (e) brain and (f) liver were enriched in intergenic and inter-CpG regions, whilst being underrepresented in CpG and gene regions. The bar plot compares the number of WGBS DMRs in red to a set of equivalent randomly generated regions in blue. (g) Gene ontology analysis of liver DMRs shows enrichment within neuronal cellular components and biological processes. BP/red point = biological process, CC/blue point = Cellular component. X-axis of point indicates FDR of ontology. Size of point indicates number of overlapping genes with ontology. There were insufficient number of DMRs identified in the brain for a gene ontology analysis.

Gene	Tissue	Intronic DMR	Width	No. CpGs	Δmeth	P-value
<i>Auts2</i>	Brain	chr5:131510296-131510465	170	5	-23.5%	<0.05
<i>Auts2</i>	Liver	chr5:131621828-131621999	172	4	-22.5%	<0.05
<i>Adgb</i>	Brain	chr10:10455557-10455883	327	4	-25.0%	<0.05
<i>Adgb</i>	Liver	chr10:10353338-10353613	276	4	-25.9%	<0.05
<i>Rbfox1</i>	Brain	chr16:6813039-6813217	179	5	-24.3%	<0.05
<i>Rbfox1</i>	Liver	chr16:6781985-6782330	346	5	-22.6%	<0.05

Table 1

Table of DMRs identified in the intronic regions of genes that contained DMRs in both the brain and liver.

Δmeth indicates the percentage change in average methylation level within the DMR with PAE compared to non-PAE mice.

Assay type	Brain DMRs	Brain randomised	Liver DMRs	Liver randomised
		regions		regions
ATAC-seq	21/78 (26.92%), P = 0.01	1/78 (1.28%), P = 0.16	53/759 (6.98%) P = 0.01	22/759 (2.90%) P = 0.31
H3K4me1	4/78 (5.13%) P = 0.03	2/78 (2.56%) P = 0.18	38/759 (5.01%) P = 0.05	35/759 (4.61%) P = 0.32
H3K27ac	9/78 (11.54%) P = 0.01	2/78 (2.56%) P = 0.74	48/759 (6.32%) P = 0.01	19/759 (2.50%) P = 0.26

Table 2.

Number and percentage of brain and liver DMRs that overlap with tissue-specific regulatory regions.

ATAC-seq, H3K4me1 and H3K27ac regions were obtained at 0 days postnatal from the ENCODE database. P-values for permutation testing using a randomisation strategy.

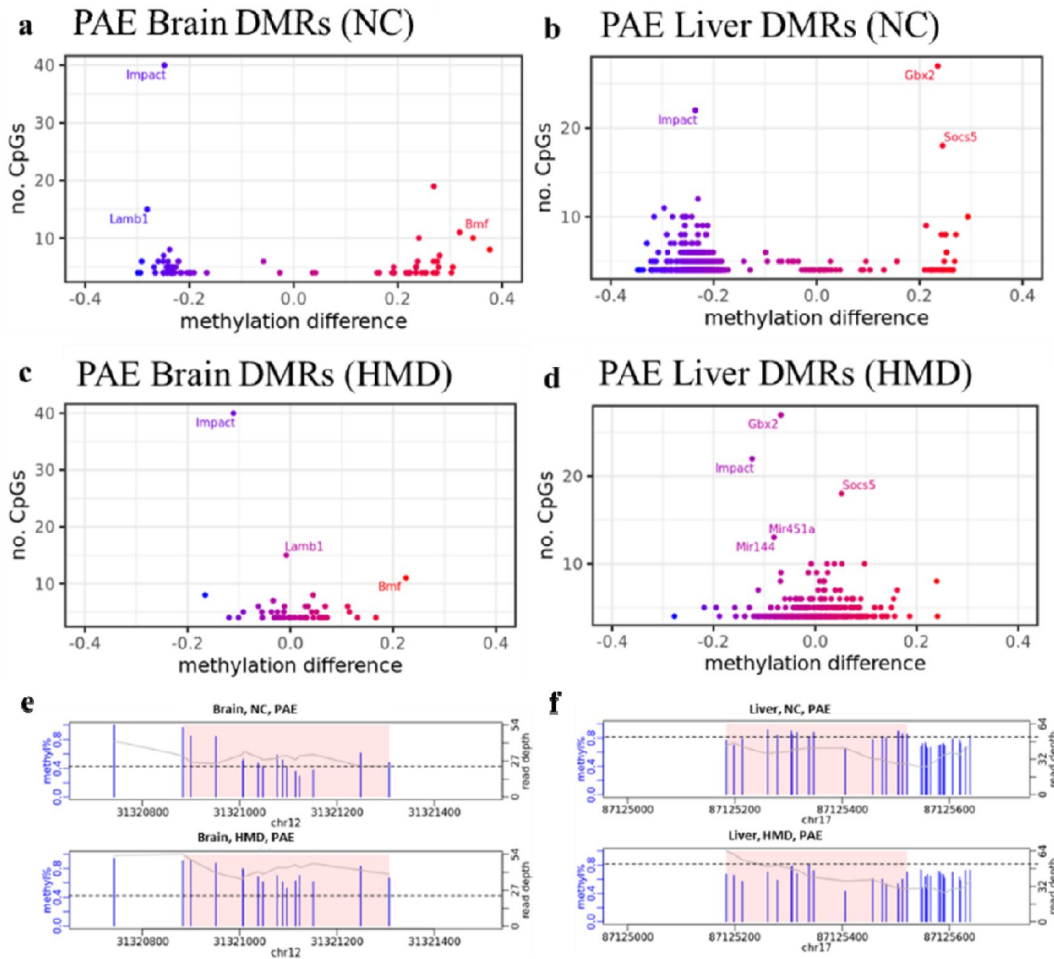


Figure 3.

HMD partially mitigated effects of PAE on offspring DNA methylation.

Average DNA methylation effect sizes above 30% with PAE were observed in some (a) brain, and (b) liver DMRs in NC mice. Mean absolute difference in methylation with PAE is reduced within the HMD mice in (c) brain, and (d) liver. Each point represents one DMR. Point colour indicates change in DNA methylation with PAE. Points with a high number of CpGs and methylation difference are annotated with associated gene if located within a genic region. HMD was associated with (e) higher methylation in the DMR identified proximal to *Lamb1* on chromosome 12 in brain and (f) lower methylation in the DMR identified proximal to *Socs5* on chromosome 17 in liver. Each plot represents a separate treatment group. Each blue vertical line indicates a CpG site, with the height and corresponding left y-axis indicating the methylation ratio. The grey line and corresponding right y-axis indicate coverage at each CpG site. The black horizontal line indicates (e) 40% and (f) 80% methylation for comparison purposes. The x-axis indicates the base position on the chromosome, with the pink shaded area highlighting the DMR. DMR plots include 200 base pair flanking regions on each side of the DMR.

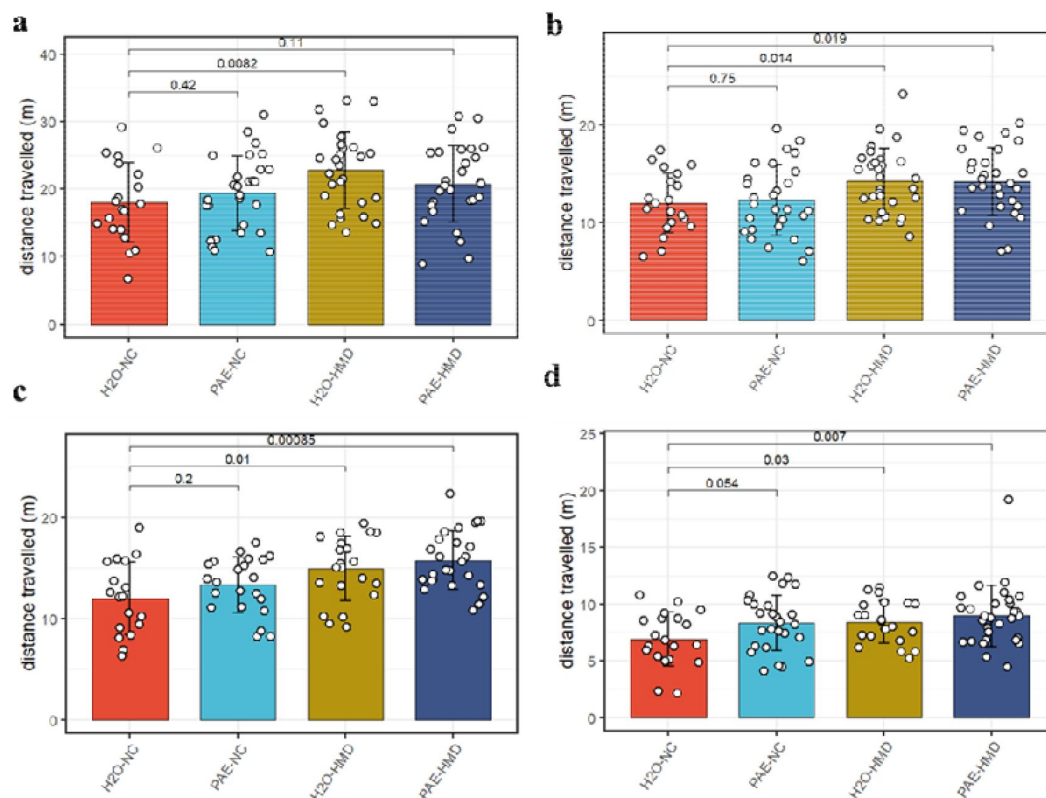


Figure 4.

HMD was associated with increased locomotor activity.

HMD was associated with increased locomotor activity compared to NC, indicated by significantly greater total distance travelled in the (a) open field test (N = 104), (b) object recognition test (N = 108), (c) elevated plus maze test (N = 88), and (d) object in place test (N = 98) by unpaired t-test. Bars show mean and standard deviation. Each point represents one mouse. NC = normal chow, HMD = high methyl diet, PAE = prenatal alcohol exposure. Time interval for each mouse was (a-c) 300 seconds and (d) 180 seconds.

Replication studies in Human PAE and FASD case-control cohorts

We undertook validation studies by examining PAE sensitive regions identified in our murine model using existing DNA methylation data from human cohorts to address the generalizability of our findings. Only 36 of the 78 (46.2%) brain DMRs, and 294 of the 759 (38.8%) liver DMRs, had homologous regions in the human genome that were able to be tested. In this validation study, DNA methylation array data from 147 newborn buccal swabs from the Asking Questions About Alcohol in Pregnancy (AQUA) cohort (5 [↗](#)) were available from this cohort (96 moderate PAE and 51 controls). We performed differential testing on a total of 1,898 CpG sites that corresponded to mouse DMRs, comparing ‘never exposed’ newborns to ‘any exposure’ and found no evidence of differential DNA methylation at these CpG (data not shown). We also accessed publicly available DNA methylation array measurements from buccal swabs taken from a Canadian clinical cohort of children with diagnosed FASD, and controls. To avoid confounding due to ancestry, we analysed the 118 Caucasian individuals (30 FASD and 88 controls). Differential testing of a total of 2,316 CpG sites that were homologous to mouse DMRs statistically replicated 7 DMR associations with FASD status (FDR $P < 0.05$) after adjusting for participant age, sex, array number, and estimated cell counts (**Table 3** [↗](#)). Visual comparisons of DNA methylation across these seven DMRs revealed striking differences in effect sizes between people with FASD and our murine model (**Figure 5** [↗](#)). Genes associated with these DMRs are linked to clinically relevant traits in the GWAS catalogue including facial morphology (*GADD45A* (26 [↗](#))), educational attainment (*AP2B1* (27 [↗](#))), intelligence (*RP9* (28 [↗](#))), autism and schizophrenia (*ZNF823* (29 [↗](#))).

Candidate Gene Analysis of previously defined alcohol sensitive regions

We also undertook a replication analysis in our murine data of previously published alcohol sensitive regions by undertaking a systematic review of previously published mammalian studies (7 [↗](#)). Candidate gene studies identified 21 CpG sites (FDR<0.05) in the brain from 15,132 CpG sites tested, including two sites in the *Mest* (*Peg1*) gene and 19 sites in *Kcnq1* (*KvDMR1*) (Table S9). There were nine FDR-significant CpG sites identified in the liver out of 15,382 CpG sites tested, all of which were in *Peg3* (Table S10). All FDR-significant CpG sites from both tissues had higher DNA methylation in mice with PAE.

Discussion

In this study, we found that moderate early (first trimester) PAE was sufficient to induce site-specific differences to DNA methylation in newborn pups without causing overt behavioural outcomes in adult mice. Although global levels of DNA methylation were not significantly different with PAE, regional analysis demonstrated widespread effects characterised predominantly by lower DNA methylation with PAE, mostly at non-coding regions of the genome. In our model, alcohol effects on DNA methylation were predominantly tissue-specific, with only two genomic regions and four genes that were similarly affected in both liver and brain. These perturbations may have been established stochastically because of PAE to the early embryo and maintained in the differentiating tissue. Further analysis in different germ layer tissues is required to formally establish this. Indeed, most of the observed effects were tissue-specific, with more perturbations to the epigenome observable in liver tissue, which may reflect the liver’s specific role in metabolic detoxification of alcohol. Alternatively, cell type composition differences between brain and liver might explain differential sensitivity to alcohols effects. Generally, DMRs were enriched in non-coding regions of the genome with regulatory potential, suggesting alcohol has broad effects on genome regulation.

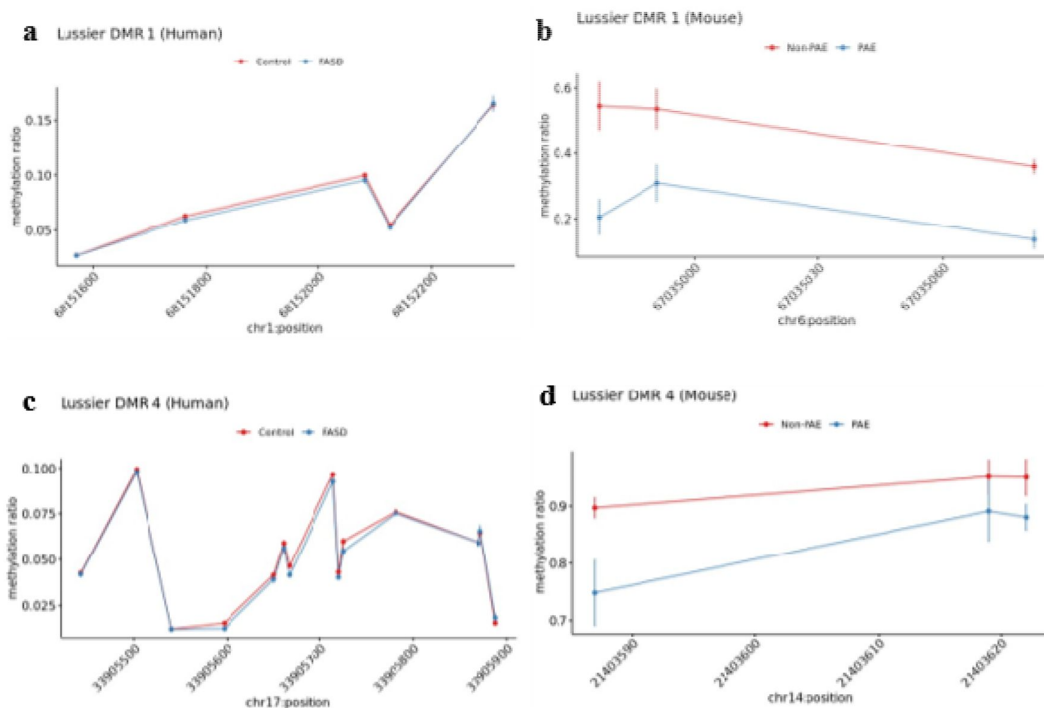


Figure 5.

Seven PAE DMRs identified in the murine model were successfully replicated in the Lussier et al. human FASD cohort.

Examples of two PAE DMRs that had significantly lower DNA methylation with a clinical diagnosis of FASD in the Lussier et al. cohort (a and c), while their mouse leftover DMR also had significantly lower DNA methylation with PAE in the murine model experiment (b and d).

DMR	organism	tissue	chr	start	end	width	no. cpgs	FDR	meandiff	gene
1	human	buccal	1	68151571	68152310	740	5	0.028636	-0.00497	<i>GADD45A</i>
2	human	buccal	19	13000782	13002357	1576	11	0.000197	-0.00203	<i>GCDH</i>
3	human	buccal	7	33148815	33149316	502	11	0.001149	-0.00011	<i>RP9</i>
4	human	buccal	17	33905444	33905888	445	14	0.000171	-0.00359	<i>AP2B1, PE X12</i>
5	human	buccal	17	27181503	27182342	840	11	0.018536	-0.00246	<i>ERAL1, FA M222B</i>
6	human	buccal	19	12992181	12992479	299	9	0.037431	-0.00179	<i>CTD- 2265O21.7, DNASE2</i>
7	human	buccal	19	11849531	11850013	483	9	0.022724	-0.00244	<i>ZNF823</i>
1	mouse	liver	6	67034885	67035082	197	4	<0.05	-0.220833	<i>E230016M1 1Rik</i>
2	mouse	liver	8	84901298	84901544	246	5	<0.05	-0.234457	<i>Klfl</i>
3	mouse	liver	9	22453836	22453893	57	5	<0.05	-0.226427	<i>Rp9</i>
4	mouse	brain	14	21403570	21403622	52	4	<0.05	-0.234193	<i>Adk</i>
5	mouse	liver	11	78069463	78070002	539	9	<0.05	-0.255864	<i>Mir144, Mir451a</i>
6	mouse	liver	11	78072079	78072313	234	4	<0.05	-0.215227	<i>Mir144, Mir451a</i>
7	mouse	liver	2	177091927	177092945	1018	5	<0.05	-0.224354	Intergenic

Table 3.

DMRs identified in the murine model that were validated in the Lussier et al. human case-control cohort for a clinical diagnosis of FASD.

The upper section describes properties of Lussier et al. human DMRs. The lower section describes properties of the equivalent murine model DMRs.

Both the human replication studies and the candidate gene analysis provide validity to our model for recapitulating some of the genomic disturbances reported in patients with clinical FASD. It is remarkable that some associations identified in our murine model of early moderate exposure were recapitulated in human subjects with FASD despite species and biosample differences, suggesting that at least some DNA methylation changes are stable over time. Notably the effect size for replicated regions were strikingly smaller in blood samples from subjects with FASD suggesting the dose and duration of exposure may need to exceed a high threshold to survive reprogramming in the blood. We speculate this may explain lack of reproducibility in the AQUA cohort.

In the candidate gene analysis we replicated previously published reports of decreased DNA methylation within *Peg3* and *KvDMR1* from South African children with fetal alcohol syndrome (11 [↗](#)). Both genes are methylated in a parent-of-origin specific manner, suggesting that alcohol may affect imprinting processes. Previous rodent and human studies have identified DNA methylation differences with PAE in imprinted regions such as the *Igf2/H19* locus (13 [↗](#), 16 [↗](#), 30 [↗](#)), although results are not entirely consistent (31 [↗](#), 32 [↗](#)). On the balance of this, we speculate duration of exposure, dose, and other tissue-related factors all likely influence the extent to which genome-regulation is perturbed and manifests as differences in DNA methylation.

Our results are encouraging for biomarker studies and aid in the prioritisation of associations for future follow-up, particularly in relation to diagnosis of FASD. For example, three genes that were validated in the Lussier et al. cohort are zinc finger proteins (*RP9*, *PEX12*, and *ZNF823*) that play an important role in fetal gene regulation. Notably, *PEX12* is associated with Zellweger syndrome, which is a rare peroxisome biogenesis disorder (the most severe variant of Peroxisome biogenesis disorder spectrum), characterised by neuronal migration defects in the brain, dysmorphic craniofacial features, profound hypotonia, neonatal seizures, and liver dysfunction (33 [↗](#)). Future studies could perform transcriptomic analysis to investigate Another key finding from this study was that HMD mitigated some of the effects of PAE on DNA methylation. Although a plausible alternative explanation is that some of the PAE regions were not reproduced in the set of mice given the folate diet, our data are consistent with preclinical studies of choline supplementation in rodent models (34 [↗](#), 35 [↗](#)) (36 [↗](#)). Moreover, a subset of PAE regions were statistically replicated in subjects with FASD, suggestive of robust associations. Although our findings should be interpreted with caution, they collectively support the notion that alcohol induced perturbation of epigenetic regulation may occur, at least in part, through disruption of the one-carbon metabolism. The most encouraging aspect of this relates to the potential utility for evidence-informed recommendations for dietary advice or supplementation, particularly in population groups with limited access to antenatal care or healthy food choices.

Strengths of this study include the use of controlled interventions coupled with comprehensive assessment of the effects of PAE on multiple tissues. We also performed whole-genome bisulfite sequencing representing the gold standard in DNA methylation analysis, which to our knowledge has not been performed before in the context of murine PAE studies. Our findings were partially generalisable in replication studies addressing the robustness of our experimental approach. Caveats of our study design include a limited ability to determine the contribution of specific cell types within tissues to the methylation differences observed, and we did not assess markers of brain or liver physiology. Additionally, we employed an ad-libitum alcohol exposure model rather than direct dosing of dams. Although the trajectories of alcohol consumption were not statistically different between groups, this introduces more variability into alcohol exposure patterns, and might impact offspring methylation data. Despite these limitations, the results were meaningful in the context of typical patterns of alcohol consumption in human populations.

In conclusion, this study demonstrates that early moderate PAE can disturb fetal genome regulation in mice and humans and supports current public health advice that alcohol consumption during pregnancy, even at low doses, may be harmful.

Materials and Methods

Murine subjects and housing

To study the effects of early moderate PAE on offspring DNA methylation processes, we adapted a murine model study design that has previously reported DNA methylation changes at the A^{VY} locus in Agouti mice (37) (Figure 1). This study received animal ethics approval from the Telethon Kids Institute Animal Ethics Committee (Approval Number: 344). Sixty nulliparous C57BL/6J female mice aged ~8 weeks were mated with equivalent stud male mice. Pregnant dams were randomly assigned to one of four treatment groups (n = 15 dams per group) that varied based on composition of the drinking water and chow given to the dams:

1. PAE-NC (Prenatal Alcohol Exposure-Normal Chow): 10% (v/v) ethanol in non-acidified water *ad libitum* from 10 days prior to mating until gestational days (GD) 8-10. This is intended to replicate typical patterns of drinking during the first trimester of pregnancy in humans. Dams received non-acidified reverse osmosis water for the remainder of pregnancy and normal chow (Rat and Mouse Cubes, Speciality Feeds, Glen Forrest, Australia) throughout pregnancy.
2. PAE-HMD (Prenatal Alcohol Exposure-High Methyl Donor diet): 10% (v/v) ethanol in non-acidified water *ad libitum* from 10 days prior to mating until GD8-10 and non-acidified reverse osmosis water for remainder of pregnancy. Isocaloric high methyl donor (HMD) chow consisting of 20 mg/kg folate and 4,970 mg/kg choline throughout pregnancy (Speciality Feeds, Glen Forrest, Australia).
3. H₂O-NC (Water-Normal Chow): non-acidified water and normal chow throughout pregnancy.
4. H₂O-HMD (Water-High Methyl Donor diet): non-acidified water and HMD chow throughout pregnancy.

Whole-genome bisulfite sequencing of newborn mouse tissues

Pups selected for WGBS in each intervention group were matched on sex and litter size to minimize variability in exposure. Two male and two female pups per treatment group (n = 16 total) were euthanised by intraperitoneal injection with ketamine and xylazine on the day of birth for WGBS of their brain and liver tissues. Mouse tissue samples were stored at -80°C. Remaining littermates grew until adulthood for behavioural testing. Ten milligrams of tissue were collected from each liver and brain and lysed in Chemagic RNA Tissue10 Kit special H96 extraction buffer. Total nucleic acid was extracted from the tissues using the Chemagic 360 instrument (PerkinElmer) and quantified with Qubit DNA High Sensitivity Kit (Catalogue Number: Q32854, Thermo Scientific). 100 ng of genomic DNA was spiked with 0.5 ng of unmethylated lambda DNA (Catalogue Number: D1521, Promega) to assess the bisulfite conversion efficiency. Each sample was digested with 2 µl RNase A (Invitrogen) at 37°C for 20 minutes to remove RNA. 100 ng of genomic DNA from each sample was sheared using a Covaris M220 (300bp settings, Covaris). Libraries were prepared using the Lucigen NxSeq AmpFREE Low DNA Library Kit (Catalogue Number: 14000-1, Lucigen), according to the manufacturer's instructions. Nextflex bisulfite-seq barcodes (Catalogue Number: Nova-511913, PerkinElmer) were used as the adapters with incubation at 25°C for 30 minutes. The libraries were bisulfite converted using the Zymo EZ DNA Methylation-Gold Kit (Catalogue Number: D5005, Zymo Research) and PCR amplified using the KAPA HiFi Uracil PCR Kit (Catalogue Number: ROC-07959052001, Kapa Biosystems). The final libraries were assessed with the Agilent 2200 TapeStation System using D1000 Kit (Catalogue Number: 5067-5582). WGBS was performed by Genomics WA sequencing core on a NovaSeq 6000 (Illumina) using 2×150bp chemistry on an S4 flow cell. The bisulfite conversion rate in each tissue sample was at least 99%. The overall mean coverage in each sample was 9.69x (range: 6.51-12.12x).

Behavioural testing in adult mice

Littermates who were not sacrificed at birth were reared on normal chow and drinking water *ad libitum* until adulthood (~8 weeks after birth) when they underwent behavioural tests assessing five neurodevelopmental domains that can be affected by PAE including locomotor activity, anxiety, spatial recognition, memory, motor coordination and balance. These tests included the open field test (locomotor activity, anxiety) (38 [↗](#)), object recognition test (locomotor activity, spatial recognition) (39 [↗](#)), object in place test (locomotor activity, spatial recognition) (40 [↗](#)), elevated plus maze test (locomotor activity, anxiety) (41 [↗](#)), and two trials of the rotarod test (motor coordination, balance) (42 [↗](#)). Between mouse subjects, behavioural testing equipment was cleaned with 70% ethanol. Video recording was employed for all behavioural tests, except for the rotarod, and the assessment process was semi-automated using ANY-maze software (Stoelting Co., Wood Dale, Illinois, U.S.A.).

Statistical analysis

Dam characteristics and pup behavioural testing results were generally assessed using unpaired t-tests comparing each treatment group to the baseline control group that was given non-acidified reverse osmosis water and normal chow throughout pregnancy. Trajectories of liquid consumption and weight gain across pregnancy, which were assessed using a quadratic mixed effects model and the trajectory of chow consumption across pregnancy which was assessed using a linear mixed effects model. To examine the effect of alcohol exposure on behavioural outcomes we used linear regression with alcohol group (binary) as the main predictor adjusted for diet and sex. For sequencing data, raw fastq files were mapped to the mm10 mouse reference genome with BSseeker 2 (version 2.1.8) (43 [↗](#)) and CG-maptools (version number 0.1.2) (44 [↗](#)) using a custom bioinformatics pipeline. CGmap output files were combined as a bsseq object in the R statistical environment (45 [↗](#)). We filtered the sex chromosomal reads and then combined reads from mice in the same treatment group using the *collapseBSseq* function, to maximise coverage prior to differential DNA methylation analysis. CpG sites with an aggregated coverage below 10x in each tissue type were removed prior to modelling to ensure there was sufficient coverage in all assessed CpG sites. This retained 94.9% of CpG sites in the brain and 93.8% of CpG sites in the liver. Differentially methylated regions (DMRs) were identified within each tissue using a Bayesian hierarchical model comparing average DNA methylation ratios in each CpG site between PAE and non-PAE mice using the Wald test with smoothing, implemented in the R package *DSS* (46 [↗](#)). False-discovery rate control was achieved through shrinkage estimation methods. We declared DMRs as those with a local FDR P-value < 0.05 based on the p-values of each individual CpG site in the DMR, and minimum mean effect size (delta) of 5%. Gene ontology analysis was performed on the brain and liver DMRs using the Gene Set Enrichment Analysis computational method (47 [↗](#)) to determine if the DMRs were associated with any transcription start sites or biological processes. Brain and liver DMRs were tested for enrichment within ENCODE Project data sets (48 [↗](#)) by an overlap permutation test with 100 permutations using the *regioneR* package. The ENCODE Project data sets that were assessed included ENCFF845WSI, ENCFF764NTQ, ENCFF937JHP, ENCFF269TLO, ENCFF676TSV and ENCFF290MLR. DMRs were then tested for enrichment within specific genic and CpG regions of the mouse genome, compared to a randomly generated set of regions in the mouse genome generated with *resampleRegions* in *regioneR*, with equivalent means and standard deviations. For candidate gene analysis, we compiled a set of key genes and genomic regions identified in previous mammalian PAE studies for site-specific testing based on our prior systematic review (7 [↗](#)), which identified 37 candidate genes (Tables S5-6). Murine brain and liver datasets were filtered to candidate gene regions and differential testing was then performed across the entire coding sequence, separately in the brain and liver of the mice on a normal diet using the *callDML* feature in *DSS*.

Validation studies in human cohorts

We used existing human data sets to validate observations from our murine model, focusing on regions identified in our early moderate PAE model. Validation studies in human cohorts with existing genome-wide DNA methylation data sets and matching PAE data are described in the Supplementary Material. Briefly, Illumina Human Methylation array .iDAT files were pre-processed using the *minfi* package (49) from the Bioconductor project (<http://www.bioconductor.org>) in the R statistical environment (<http://cran.r-project.org/>, version 4.2.2). Sample quality was assessed using control probes on the array. Between-array normalization was performed using the stratified quantile method to correct for Type 1 and Type 2 probe bias. Probes exhibiting a *P*-detection call rate of >0.01 in one or more samples were removed prior to analysis. Probes containing SNPs at the single base extension site, or at the CpG assay site were removed, as were probes measuring non-CpG loci (32,445 probes). Probes reported to have off-target effects in McCartney et al. (50) were also removed. Mouse DMRs were converted into human equivalent regions using an mm10 to hg19 genome conversion with the liftOver tool in the UCSC Genome Browser (51). A minimum 0.1 ratio of bases that must remap was specified as recommended for liftOver between regions from different species and multiple output regions were allowed. Differential testing of candidate mouse DMRs was carried out using the R package *DMRcate* (52) for each dataset and DMRs were declared as minimum smoothed false-discovery rate (FDR) < 0.05. Cell heterogeneity in each sample including the composition of epithelial, fibroblast and immune cells was estimated from DNA methylation reads using the R package *EpiDISH* (53).

Data Availability Statement

Data and code pertaining to the original work in this article are available upon request from the corresponding author.

Conflict of Interest Statement

The authors declare no conflicts of interest.

Author Contributions

D. Martino, M. Symons, A. Larcombe, R. Lister, D. Hutchinson, E. Muggli, J. Craig, J. Halliday, J. Fitzpatrick, S. Buckberry, M. Bestry and E. Elliott contributed to the study design and funding application. E. Chivers, A. Larcombe performed the mouse work including administering the mating, treatments, measurements, monitoring and extracting biological samples. E. Chivers, A. Larcombe and M. Bestry performed the mouse behavioural testing. C. Bakker analysed the videos from the mouse behavioural testing. M. Bestry prepared the whole-genome bisulfite sequencing libraries and performed the data analyses. N. Kresoje assisted with preparing whole-genome bisulfite sequencing libraries. S. Buckberry and D. Martino provided advice and support on the data analysis. J. Halliday and E. Muggli contributed human datasets for reproducibility analysis. M. Bestry and D. Martino drafted the manuscript. M. Symons, A. Larcombe, R. Lister, D. Hutchinson, E. Muggli, J. Craig, J. Halliday, J. Fitzpatrick, S. Buckberry, E. Elliott and N. Kresoje contributed to the development and editing of the manuscript.

Acknowledgements

We wish to acknowledge the assistance of Dr Jahnvi Pflueger who provided training on preparation of whole-genome bisulfite sequencing libraries. We wish to acknowledge the financial contribution of the Centre for Research Excellence in FASD who supported the murine experiments.

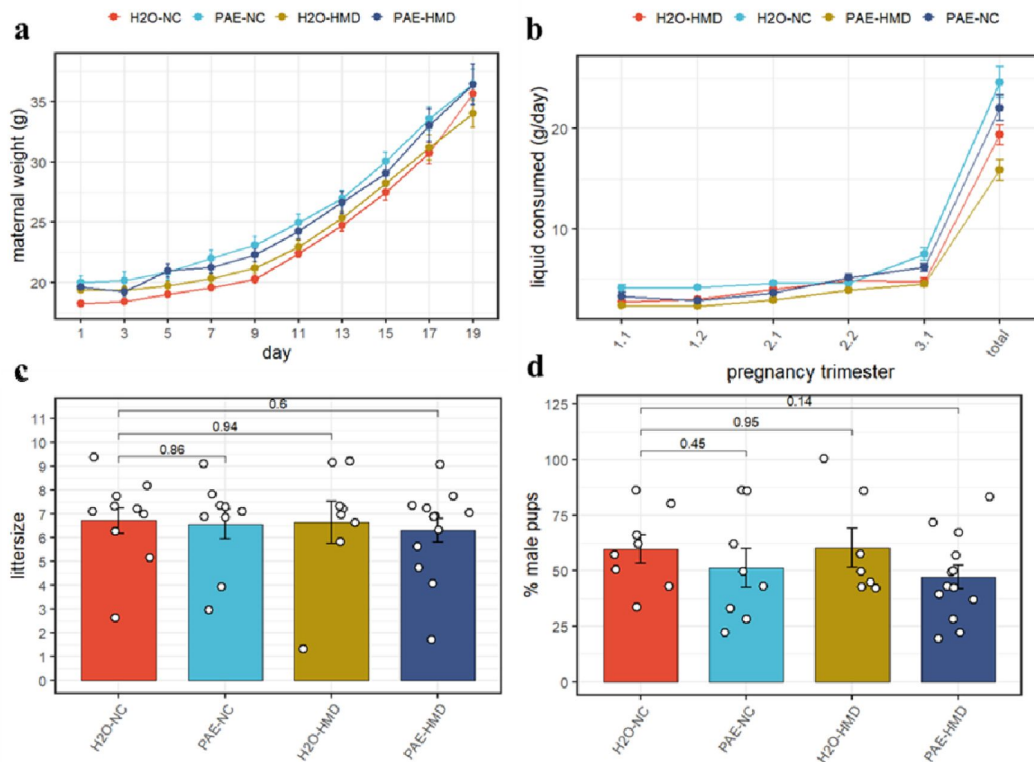


Figure S1.

PAE and HMD effects on dam characteristics.

(a) Dam weight progression was significantly affected by HMD but not PAE by quadratic mixed-effects model without interaction (b) Trajectory of liquid consumption across pregnancy was affected by PAE and HMD by quadratic mixed effects model. PAE and HMD significantly interacted with trimester of pregnancy. (c) litter size (N = 40) and (d) pup sex ratios (N = 36) were not significantly associated with PAE or HMD by unpaired t-test or ANOVA. All line and bar plots show mean and standard deviation. NC = normal chow, HMD = high methyl diet, PAE = prenatal alcohol exposure. Comparisons show p-value by unpaired t-test compared to the H₂O-NC baseline treatment group.

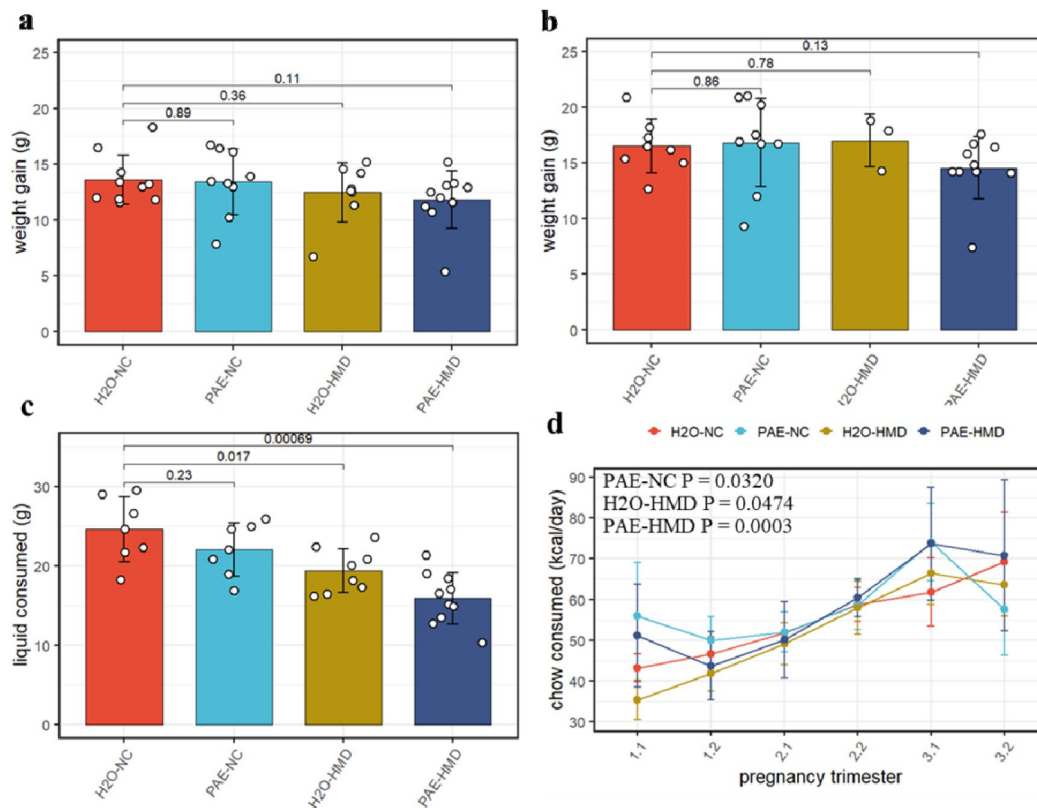


Figure S2.

PAE and HMD effects on dam characteristics.

There was no significant difference in the average gain of weight in dams between (a) days 1-17 or (b) days 1-19 by treatment group. Both timepoints were included due to some pregnancies ending by day 19. (c) Dams given supplemented chow consumed significantly lower total quantity of liquid across pregnancy. Bar plots show mean and standard deviation for each treatment group. Each point represents one dam. (d) the trajectory of chow consumed by dams across pregnancy significantly varied with the addition of treatments. Points show mean and standard deviation for each treatment group. Statistical analysis involved linear mixed-effects regression comparing trajectories of treatment groups to H2O-NC baseline control group.

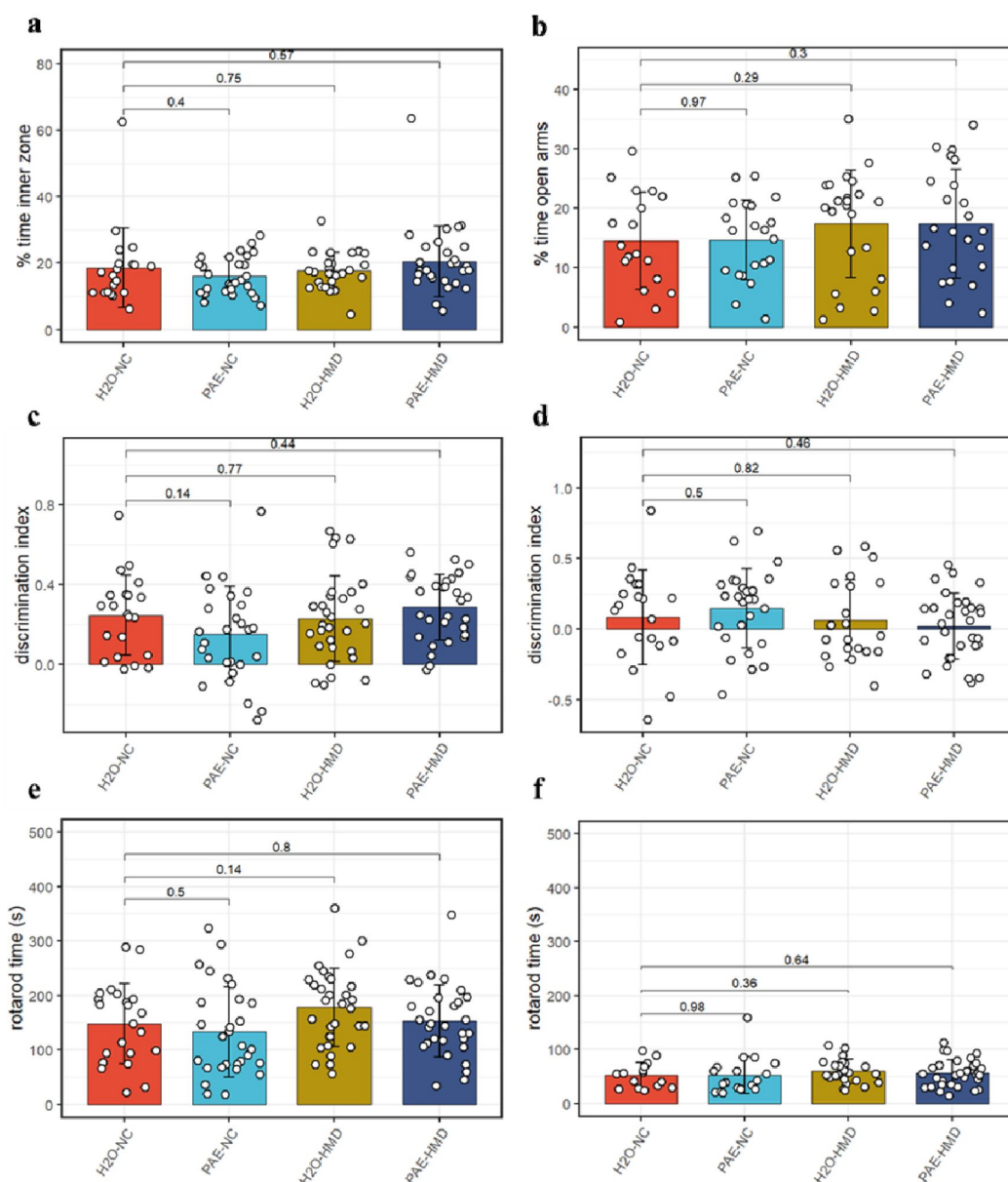


Figure S3.

PAE had no significant effect on other assessed behavioural outcomes.

PAE and HMD had no significant effect on anxiety as evident by no significant difference by unpaired t-test in the (a) percent time in the inner zone in the open field test (N = 104) and (b) percent time open arms in the elevated plus maze test (N = 85). PAE and HMD had no significant effect on spatial recognition as evident by no significant difference by unpaired t-test in the discrimination index in (c) object recognition (N = 108) and (d) object in place test (N = 98). PAE and HMD had no significant effect on motor co-ordination and balance as evident by no significant difference by unpaired t-test in times in (e) first rotarod test (N = 112) and (f) second rotarod test (N = 87). Bars show mean and standard deviation. Each point represents one mouse. NC = normal chow, HMD = high methyl diet, PAE = prenatal alcohol exposure. Time interval for each mouse was (a-c) 300 seconds and (d) 180 seconds.

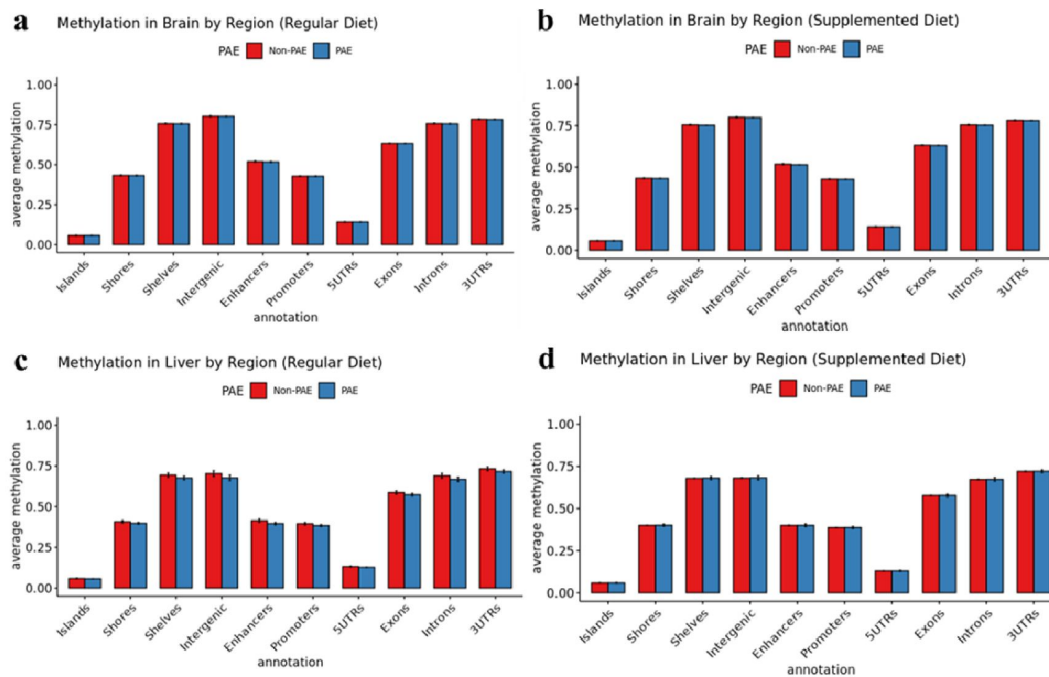


Figure S4.

No evidence for global disruption of methylation by PAE.

The figure shows methylation levels averaged across CpGs in different regulatory genomic contexts. Neither brain tissue (A & B), nor liver tissue (C & D) were grossly affected by PAE exposure (blue bars). Bars represent means and standard deviation.

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Reviewer #1 (Public Review):

Summary:

This manuscript examined the impact of prenatal alcohol exposure on genome-wide DNA methylation in the brain and liver, comparing ethanol-exposed mice to unexposed controls. They also investigated whether a high-methyl diet (HMD) could prevent the DNA methylation alterations caused by alcohol. Using bisulfite sequencing (n=4 per group), they identified 78 alcohol-associated differentially methylated regions (DMRs) in the brain and 759 DMRs in the liver, of which 85% and 84% were mitigated by the HMD group, respectively. The authors further validated 7 DMRs in humans using previously published data from a Canadian cohort of children with FASD.

Overall, the findings from this study provide new insight into the impact of prenatal alcohol exposure, while also showing evidence for methyl-rich diets as an intervention to prevent the effects of alcohol on the epigenome. Some methodological concerns and confounders limit the robustness of these results, and should be addressed in future studies to further strengthen the conclusions of this study and its applicability to broader settings.

Strengths:

- The use of whole genome bisulfite sequencing allowed for the interrogation of the entire DNA methylome and DMR analysis, rather than a subset of CpGs.
- The combination of data from animal models and humans allowed the authors to make stronger inferences regarding their findings

- The authors investigated a potential mechanism (high methyl diet) to buffer against the effects of prenatal alcohol exposure, which increases the relevance and applicability of this research.

Weaknesses:

- The sample size was small for the epigenetic analyses, which limits the robustness of the findings.
- The authors could not account for potential confounders in their analyses, including birthweight, alcohol levels, and sex. This is a particular problem for the high-methyl diet analyses, in which the alcohol-exposed mice consumed less alcohol than their non-diet counterparts.

<https://doi.org/10.7554/eLife.92135.2.sa1>

Reviewer #2 (Public Review):

Summary:

Bestry et al. investigated the effects of prenatal alcohol exposure (PAE) and high methyl donor diet (HMD) on offspring DNA methylation and behavioral outcomes using a mouse model that mimics common patterns of alcohol consumption in pregnancy in humans. The researchers employed whole-genome bisulfite sequencing (WGBS) for unbiased assessment of the epigenome in the newborn brain and liver, two organs affected by ethanol, to explore tissue-specific effects and to determine any "tissue-agnostic" effects that may have arisen prior to the germ-layer commitment during early gastrulation. The authors found that PAE induces measurable changes in offspring DNA methylation. DNA methylation changes induced by PAE coincide with non-coding regions, including enhancers and promoters, with the potential to regulate gene expression. Though the majority of the alcohol-sensitive differentially methylated regions (DMRs) were not conserved in humans, the ones that were conserved were associated with clinically relevant traits such as facial morphology, educational attainment, intelligence, autism, and schizophrenia. Finally, the study provides evidence that maternal dietary support with methyl donors alleviates the effects of PAE on DNA methylation, suggesting a potential prenatal care option.

Strengths:

The strengths of the study include the use of a mouse model where confounding factors such as genetic background and diet can be well controlled. The study performed whole-genome bisulfite sequencing, which allows a comprehensive analysis of the effects of PAE on DNA methylation.

Weaknesses:

Transcriptome analysis to test if the identified DMRs indeed affect gene expression would help determine the potential function of the identified methylation changes.

<https://doi.org/10.7554/eLife.92135.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Response to reviewers

We wish to thank the reviewers for the time taken to appraise the manuscript and the helpful feedback to improve it. We have taken onboard the suggested feedback and incorporated it into the revision. The findings of the revised manuscript are unchanged. Below is a point-by-point response to specific comments.

Public reviews

Reviewer 1

Thank you to reviewer 1 for the thorough and insightful review of our manuscript. We are pleased that the strengths of our research, particularly the use of whole-genome bisulfite sequencing, the combination of animal and human data, and the investigation of a potential dietary intervention were recognized. We are confident that these aspects contribute significantly to the value and originality of our work.

We acknowledge the concerns regarding the statistical rigor of the study, particularly the sample size and data analysis methods. We would like to address these points in more detail:

Sample size: While we agree that a larger sample size would be ideal, the chosen sample size (n=4 per group) is consistent with other murine whole-genome bisulfite sequencing experiments in the field. We have carefully considered the cost-benefit trade-off in selecting this approach. In the revision we discuss the potential limitations of this sample size.

Data analysis: We acknowledge the inconsistencies in the study reporting and have committed to improving the clarity in the revision. We carefully reviewed the concerns regarding the use of causal language and the interpretation of differences in our results. In some cases, the use of causal language is justified by the intervention study design. We also believe other explanations like stochastic variation affecting the same genomic regions in different tissues, are exceedingly unlikely from a statistical viewpoint. In the revision we have adopted a balanced approach to the language.

Confounders: We acknowledge the importance of accounting for potential confounders such as birthweight, alcohol exposure and sex. The pups selected for genome analysis were matched for sex and on litter size as a proxy for in utero alcohol exposure. This careful selection of mice for genome analysis was intentionally guided to mitigate potential confounding.

Statistical rigour: We acknowledge the importance of multiple testing correction in the genome-wide analysis. We used the DSS method of Feng et al (PMID: 2456180) which employs a two-step procedure for assessing significance of a region. Instead of a single p-value for the whole DMR, DSS uses the area statistic to rank candidate regions and control the false discovery rate through shrinkage estimation methods. This approach reduces the risk of reporting false positives due to multiple testing across numerous CpG sites. It is similar in respects to employing local FDR correction at 0.05 level, with an additional minimum effect size threshold applied, and particularly suited to experiments where the number of replicates is low. In the revision we have committed to improving the clarity of the reporting of statistical methods.

Thank you to reviewer 2 for the comprehensive and valuable feedback on our manuscript. We take your concerns about the generalizability of our findings and the interpretation of certain results seriously. We would like to address your specific criticisms in detail:

Generalizability and Human Data: We agree that the generalizability of mouse models to human conditions has limitations. However, our study focused on understanding the early

Reviewer 2

Thank you to reviewer 2 for the comprehensive and valuable feedback on our manuscript. We take your concerns about the generalizability of our findings and the interpretation of certain results seriously. We would like to address your specific criticisms in detail:

Generalizability and Human Data: We agree that the generalizability of mouse models to human conditions has limitations. However, our study focused on understanding the early molecular alterations caused by moderate PAE, which can be more effectively modelled in a controlled environment like mice. To clarify this, we have strengthened the manuscript by emphasizing the focus on moderate PAE in the title and throughout the paper.

Transcriptome Analysis: We recognize the importance of investigating the functional consequences of PAE-induced DMRs and agree that transcriptome analysis would be highly valuable. We are currently planning to conduct future transcriptomic studies to understand the link between DMRs and gene expression.

Species-Specificity and DMR Enrichment: We acknowledge the likelihood of species-specific PAE effects. Our finding of enrichment of DMRs in non-coding regions was consistent with observations from the Lussier study of FASD. We agree there is further work to do and now highlight this in the discussion.

Tissue Sample Locations: Due to technical restrictions of processing newborn mouse tissue, we are unable to enhance the manuscript with specific tissue regions sampled.

Interpretation of Shared Genomic Regions: We appreciate your point about the alternative explanation for the shared genomic regions between brain and liver. Our interpretation is that regions identified in the alcohol group only affected equally in both tissues are likely established stochastically (as a result of the exposure) in the early embryo and then maintained in the germ layers. We have revised to suggest this is the most likely explanation and we acknowledge a more detailed examination in more tissues would be warranted for proof.

Additional Feedback**Reviewer 1***Introduction*

- *Line 65 - alcohol consumption is not always preventable and these statements further increase the stigma associated with FASD. A better way to say this would be "a leading cause of neurodevelopmental impairments".*

We have implemented this suggestion in revised manuscript.

- *The studies cited in lines 87-89 are somewhat outdated, as several more recent studies with better sample sizes have been published in recent years. I would recommend citing more recent publications in addition to these studies. Similarly, the authors should also cite Portales-Casamar et al., 2016 (Epigenetic & Chromatin) for the validation in humans, as it was the original study for those data.*

We have added a citation for the study mentioned by Portales-Casamar et al. (2016) in the revised manuscript.

- *Lines 95-95 - the authors should elaborate further on the "encouraging results" from choline supplementation studies, as these details may help interpret the findings from their own study.*

In the revised manuscript, we replaced “encouraging results” with “results suggesting a high methyl donor diet (HMD) could at least partially mitigate the adverse effects of PAE on various behavioural outcomes”.

- *Minor point: DNA methylation is preferable to "methylation" alone when not referring to specific CpGs or sites, as methylation can also refer to protein or RNA methylation.*

“Methylation” has been replaced with “DNA methylation” in revised manuscript

Results

- *Line 118 - HMD should be defined here.*

HMD defined in revised manuscript

- *The figures in the main manuscript and supplemental materials are not in the same order as they are presented in the text.*

We apologise for this and thank the reviewer for their attention to detail. In the revision we have corrected the order of figures to match the text.

- *It is concerning that the H20-HMD group had lower baseline weights, which could impact the findings from these analyses. Please discuss how these differences were accounted for in the study design and analyses.*

We appreciate the reviewer's concern about the lower baseline weight in the H20-HMD group. We agree that this difference could potentially affect our findings. However, we want to emphasize that total weight gain during pregnancy was statistically similar across all groups by linear mixed effect model. Additionally, all dams were within the healthy weight range for their strain. While we cannot completely rule out any potential influence of baseline weight, we believe the similarity in weight gain and the healthy range of all dams suggest that the in-utero experience of pups regarding weight-related factors was likely comparable across groups.

- *I have some concerns regarding the cutoffs used to identify the DMRs, particularly given the small N and number of tests. The authors should report the number of DMRs that meet a multiple testing threshold; if none, they should use a more stringent threshold than $p < 0.05$, as one would expect 950,000 CpGs to meet that threshold by chance ($19,000,000 \text{ CpGs} \times 0.05$). The authors should also report the number of DMRs tested, as this will be a more appropriate benchmark for their analyses than the number of CpGs (they should also report the specific number here).*

We appreciate the reviewer's concerns regarding the DMR cut-offs. We agree that clarifying the methods and justifying our choices is crucial. Our implementation of the DSS method for defining DMRs employs a local FDR $p < 0.05$ cut-off, with additional delta beta threshold of 5%.

We have clarified this in the methods section of the revised manuscript . We want to emphasize that the local FDR approach effectively mitigates the concern of chance findings by adjusting for multiple comparisons across the genome. Line 414-420 in the revised methods contains the following amended text

“Differentially methylated regions (DMRs) were identified within each tissue using a Bayesian hierarchical model comparing average DNA methylation ratios in each CpG site between PAE and non-PAE mice using the Wald test with smoothing, implemented in the R package DSS (46). False-discovery rate control was achieved through shrinkage estimation methods. We declared DMRs as those with a local FDR P-value < 0.05 based on the p-values of each individual CpG site in the DMR, and minimum mean effect size (delta) of 5%”

- *I also have concerns about the delta cutoff for their DMRs. First, it is not clear if this cutoff is set for a single CpG or across the DMR (even then, it is not clear if this is a mean, median, max, min, etc.) Second, since the authors analyzed CpGs with 10X coverage, they can only reliably detect a delta of 0.1 (1/10 reads).*

Thank you for raising this important point. In the revision we have clarified the effect size cutoff reflects the mean effect across CpGs within the DMR as follows (line 418)

“We declared DMRs as those with a local FDR P-value < 0.05 based on the p-values of each individual CpG site in the DMR, and minimum mean effect size (delta) of 5%”

We chose the mean as it provides a comprehensive representation of the overall methylation change within the region, while ensuring all individual CpGs used in the analysis had at least 10x coverage. It is not true that we can only detect a delta of 1/10 reads, the mean effect is the relative difference in means between groups and is not dependent on the underlying sequencing depth.

- *Prenatal alcohol exposure is known to impact cell type proportions in the brain, which could lead to differences in DNAm patterns. The authors should address this possibility in the discussion, as well as examine their list of DMRs to determine if they are associated with specific brain cell types. The possibility of cell type differences in the liver should also be discussed.*

We agree with the reviewer that PAE-induced alterations in cell type proportions can influence DNA methylation patterns. While isolating specific cell types in our current study's brain and liver samples was not achievable due to tissue limitations, we acknowledge this as a limitation and recognize the need for further investigations incorporating single-cell or cell type-specific approaches in the discussion.

- *It is interesting, but maybe not surprising, that more DMRs were identified in the liver compared to the brain. This finding would warrant some additional interpretation in the discussion.*

We appreciate and agree that this finding indeed warrants further interpretation. We have added the following sentence into the discussion section of the revised manuscript that provides some potential factors behind this observation.

Lines 263 “Indeed, most of the observed effects were tissue-specific, with more perturbations to the epigenome observable in liver tissue, which may reflect the liver’s specific role in metabolic detoxification of alcohol. Alternatively, cell type composition differences between brain and liver might explain differential sensitivity to alcohols effects”.

- *Lines 148-149 - I disagree about the enrichment of decreased DNAm in brain DMRs, as 52.6% is essentially random chance. The authors should also include a statistical test here, such as a chi-squared test, to support this statement.*

We agree that a revised interpretation is warranted. The updated manuscript has been amended as follows: “Lower DNA methylation with early moderate PAE in NC mice was more frequently observed in liver DMRs (93.5% of liver DMRs), while brain DMRs were almost equally divided between lower and higher DNA methylation with early moderate PAE (52.6% of brain DMRs had lower DNA methylation with early moderate PAE).”

- *Similarly, I would recommend the authors use increased/decreased DNAm, rather than hypermethylated/hypomethylation, as the latter terms are better suited to DNAm values near 100% or 0%.*

The use of hyper/hypo methylation is still considered common and well understood even for moderate changes. We agree the use of increased/decreased is more inclusive for a broader audience, so we have amended all references accordingly in the main text.

- *Lines 153-155 - please report the statistics to support these enrichment results. A permutation test would be well suited to this analysis.*

The reporting of statistics related to the enrichment test has now been amended to read “Overlap permutation tests showed liver DMRs were enriched in inter-CpG regions and non-coding intergenic regions ($p < 0.05$), while being depleted in all CpG regions and genic regions except 1to5kb, 3UTR and 5UTR regions, where there was no significant difference (Figure 2f).”

- *Line 156 - "overwhelming enrichment" is a very strong statement considering the numbers themselves.*

Omitted “overwhelming” in revised manuscript. Revised manuscript states: “Using open chromatin assay and histone modification datasets from the ENCODE project, we found enrichment ($p < 0.05$) of DMRs in open chromatin regions (ATAC-seq), enhancer regions (H3K4me1), and active gene promoter regions (H3K27ac), in mouse fetal forebrain tissue and fetal liver (Table 2).”

- *Lines 165-167 - Please describe the analyses and metrics used to determine if the DNAm differences were mitigated in the HMD groups. As it stands, it is not clear if they are simply not significant, or if the delta was decreased. In terms of a figure, a scatter plot of the deltas for these DMRs would be better suited to visualizing these changes.*

To determine whether DMRs were mitigated we simply applied the same statistical testing procedure on the subset of PAE DMRs in the group of mice exposed to the HM diet. The sample size is the same, and the burden on multiple testing is reduced as we did not test the entire genome. We believe our interpretation stands although we have urged caution in the discussion as follows (line 319)

“Another key finding from this study was that HMD mitigated some of the effects of PAE on DNA methylation. Although a plausible alternative explanation is that some of the PAE regions were not reproduced in the set of mice given the folate diet, our data are consistent with preclinical studies of choline supplementation in rodent models (34, 35) (36). Moreover,

a subset of PAE regions were statistically replicated in subjects with FASD, suggestive or robust associations. Although our findings should be interpreted with caution, they collectively support the notion that alcohol induced perturbation of epigenetic regulation may occur, at least in part, through disruption of the one-carbon metabolism.”

- *Given the lenient threshold to identify DMRs, it is possible that PAE-associated DMRs are simply false positives and do not "replicate" in a different subset of animals. One way to check this would be to determine whether there are any differences between mitigated/unmitigated DMRs and the strength of their initial associations. Should the mitigated DMRs skew towards higher p-values and lower deltas, one might consider that these findings could be false positives.*

We appreciate the reviewer's concern about potential false positives due to the chosen DMR identification threshold. We reiterate the DMR calling thresholds were adjusted for local FDR; however, we acknowledge the need for further validation. We haven't observed this trend of mitigated DMRs having higher p-values and lower deltas, but we have replicated some PAE DMRs in independent human datasets and found support for their biological plausibility in the context of PAE.

- *Related to the HMD analyses, I am concerned that the EtOH-HMD group consumed less alcohol, which could manifest in the PAE-induced DMRs disappearing, unrelated to the HMD exposure. The authors should comment on whether the pups were matched for ethanol exposure and include sensitivity analyses that include ethanol level as a covariate to confirm that their results are not simply due to decreased alcohol exposure.*

We appreciate the reviewer's concern regarding the lower alcohol consumption by Dams in the EtOH-HMD group and its potential impact on DMRs. We agree that consistent in utero exposure is crucial for reliable results. Our pup selection for genomic analysis involved matching litter size as a proxy for in utero exposure, so even through the average alcohol consumption was lower for the EtOH-HMD group, we matched pups across treatment groups based on litter size as a proxy for alcohol intake levels, excluding pups with significantly different exposure levels. We agree more robust methods including direct measurement of blood alcohol content would improve the study. We have now incorporated this into the discussion of the revised manuscript on lines 351: “Additionally, we employed an ad-libitum alcohol exposure model rather than direct dosing of dams. Although the trajectories of alcohol consumption were not statistically different between groups, this introduces more variability into alcohol exposure patterns, and might impact offspring methylation data”

- *Lines 172 - please be more specific about the neurocognitive domains tested.*

In the revision we have included more detail about the neurocognitive domains tested (originally mentioned in the results) in the methods as follows:

“These tests included the open field test (locomotor activity, anxiety) (38), object recognition test (locomotor activity, spatial recognition) (39), object in place test (locomotor activity, spatial recognition) (40), elevated plus maze test (locomotor activity, anxiety) (41), and two trials of the rotarod test (motor coordination, balance) (42)”

- *Line 191 - please report the tissue type used in the human study, as well as the method used to estimate cell type proportions.*

We stated in the results section that buccal swabs were used in both human cohorts.

We added to the revised manuscript that cell type proportions were estimated using the EpiDISH R package.

- *Related to validation, it is unclear whether the human-identified DMRs were also validated in mice, or if the authors are showing their own DMRs. Please also discuss why DMRs might not have been replicated in AQUA.*

We used human data sets to validate observations from our murine model, focusing on regions identified in our early moderate PAE model. This is now explicitly state on line 209 of the revision:

“We undertook validation studies by examining PAE sensitive regions identified in our murine model using existing DNA methylation data from human cohorts to address the generalizability of our findings.”

“In the section entitled ‘Candidate Gene Analysis.’ we used our murine data sets to reproduce previously published associations that included regions identified in both animal and human studies. We posit the lack of replication of our early moderate PAE regions in AQUA is explained in part by species-specific differences and considering the striking differences in effect size seen in regions that did replicate in FASD subjects, the exposure may need to be of sufficient magnitude and duration for the effects seen in brain and liver to survive reprogramming in the blood. The AQUA cohort is largely enriched for low to moderate patterns of alcohol consumption.

- *Line 197 - please provide a citation for the ethanol-sensitive regions. There are also several existing DNAm analyses in brain tissues from animal models that should be included as part of these analyses, as several have shown brain-region and sex-specific DMRs related to prenatal alcohol exposure. These contrasts might help the authors further delineate the effects of prenatal alcohol in their model and expand on current literature to explain the deficits caused by alcohol exposure.*

Our candidate gene/region selection was informed by a systematic review of previously published human and animal studies reporting associations between in utero exposure to PAE and offspring DNA methylation. We synthesized evidence across several models, tissues and methylation platforms to arrive at a core set of reproducible associations. Line 481 of the methods now includes a citation to our systematic review which details our selection criteria.

Discussion

- *Line 211 - This is a strong statement for one hypothesis. It is also possible that different cell types have similar responses to prenatal alcohol exposure. In this scenario, perturbations need not arise before germ layer separation. The authors should soften this causal statement.*

We appreciate this point although given the genome size relative to the size of the DMRs we have detected, the chance that different cell types would respond similarly in exactly the same regions seems exceedingly rare. We posit a more likely explanation is early perturbations in the embryo are established stochastically as a result of the exposure (supported by the interventional design) and maintained in the differentiating tissues. We

agree further work is needed to prove this, specifically in a wider set of tissues from multiple germ layers so we have amended the discussion as follows:

“These perturbations may have been established stochastically because of alcohol exposure in the early embryo and maintained in the differentiating tissue. Further analysis in different germ layer tissues is required to formally establish this.”

- *Lines 222-224 - I completely agree with this statement. However, the authors had the opportunity to examine dosage effects in their model as they measured alcohol-levels from the dams. At the very least, I would recommend sensitivity analyses in their DMRs to assess whether alcohol level/dosage influences their results.*

Although a great suggestion to improve the manuscript, we did not have opportunity to examine dosages by design as we selected mice for genome analysis with matched exposure patterns. It would be fascinating to conduct a sensitivity analysis.

Methods:

- *Please include the lysis protocol.*

Thank you for picking up this error in our reporting. We have now included the following details in the methods which improve the reproducibility of this study: “Ten milligrams of tissue were collected from each liver and brain and lysed in Chemagic RNA Tissue10 Kit special H96 extraction buffer”.

- *Please include the total reads for each sample and details of the QC pipeline, including filtering flags, quality metrics, and genome build.*

Thank you for suggesting improvements to our reporting which improve the reproducibility of this study. We have included a new supplementary tableTab of sequencing statistics and details of the quality metrics. Please note the genome build is explicitly stated in the methods already.

- *Please make your code publicly available to ensure that these analyses can be replicated.*

Thank you for this suggestion. A data availability statement has now been included in the revision and code will be made available upon request

- *Why were Y chromosome reads included in the dataset?*

Y chromosomal reads were not included in the DMR analysis. Amended “We filtered the X chromosomal reads” to “We filtered the sex chromosomal reads” in revised manuscript.

- *Please provide the number of total CpGs available for analysis.*

Added sentence into results section of revised manuscript: “A total of 21,842,961 CpG sites were initially available for analysis.” We also clarified that the ~19,000,000 CpGs were analysed following coverage filtering.

- *Please provide the parameters for the DMR analysis and report how the p-values and deltas were calculated.*

We have addressed this in previous comments

- *The supplemental materials for the human data are missing.*

Thank you for picking up this oversight. The revision now includes an additional data supplement which details the analysis of the human data sets for interested readers.

Tables and figures

- *Table 1. It is not clear how the DMRs for this table were selected. The exact p-values and FDR should also be reported in this table. The number of CpGs in these DMRS should also be reported.*

Table 1 includes select DMRs that were consistently detected in both brain and liver tissue. These are particularly of interest as they represent regions highly sensitive to alcohol exposure. We agree that exact reporting of p-values would be ideal. Instead of a single p-value for the whole DMR, DSS uses the area statistic to rank candidate regions and control the false discovery rate (FDR) through shrinkage estimation methods. In the revision we have now included region size and number of CpGs in table 1.

- *Table 3. Please include p-values for the DMR analyses.*

As above we report the area-statistic which is an equivalent measure to assess evidence for differential methylation.

- *Figure 2 (Figure 4 in revised manuscript). Please report the N for these analyses. It also seems that the pairwise t-tests were only compared to the H2O-NC, which does not provide much insight into the PAE group. The relevance of the sexP analysis to the present manuscript is also unclear.*

Figure 2 is now Figure 4 in the revision and the sample size has been included in figure legend. We compared all groups to the control group (H2O-NC) as we aimed to determine any differences in intervention groups from the control.

We apologies for lack of clarity around the ‘sex P’ terminology. This refers to the p-value for the main effect of sex on the behavioural outcome. We agree it lacks relevance since the regression models were adjusted for sex. In the revision we have updated the methods as follows (line426) and removed references to sex P

“To examine the effect of alcohol exposure on behavioural outcomes we used linear regression with alcohol group (binary) as the main predictor adjusted for diet and sex.”

- *Figure 3ef (Figure 2ef in revised manuscript). It is unclear how the regions random regions were generated. A permutation test would be relevant to determine whether there are any actual enrichment differences.*

As stated in methods section: “DMRs were then tested for enrichment within specific genic and CpG regions of the mouse genome, compared to a randomly generated set of regions in

the mouse genome generated with `resampleRegions` in `regionER`, with equivalent means and standard deviations.”

- *Figure 5. Please include the gene names for these DMRs, as well as their genomic locations. It would also be relevant to annotate these plots with the max, min, and mean delta between groups.*

Thank you, we considered this however the DMRs are not in genes so we cannot apply a gene label. The locations are reported on the x-axis and the statistics are shown in Table 3.

- *Figure S1b and S2c- It is quite worrisome that the PAE-HMD group drank less throughout pregnancy than their PAE counterparts. Please discuss how this was addressed in the analyses.*

We appreciate the reviewer's concern regarding the lower alcohol consumption in the PAE-HMD group and its potential impact on DMRs. We agree that consistent in-utero exposure is crucial for reliable results. Although the total amount of liquid consumed over pregnancy was lower in this group, they started with a lower baseline and the trajectory was not statistically different compared to other groups.

We have now incorporated this into the discussion section of the revised manuscript on lines 336: “Additionally, we employed an ad-libitum alcohol exposure model rather than direct dosing of dams. Although the trajectories of alcohol consumption were not statistically different between groups, this introduces more variability into alcohol exposure patterns, and might impact offspring methylation data.”

- *Figure S1cd. See my comments about Figure 2.*

Suggested changes have been incorporated.

- *Figure S2d. it is not clear to what the statistics presented in this panel refer. Please clarify and discuss the implications of dietary intake differences on your findings.*

Added sentence to caption in revised manuscript: “Statistical analysis involved linear mixed-effects regression comparing trajectories of treatment groups to H2O-NC baseline control group.”

- *Figure S3. See my comments about Figure 2.*

Suggested changes have been incorporated

- *Figure S4. I am confused by the color legend, as it seems both colors are PAE. I also do not see how any regions show increased or decreased DNAm in PAE based on this plot (also no statistics are presented to support these conclusions).*

The plot is intended to show there are no gross changes in methylation when averaged across all CpGs within different regulatory genomic contexts. Statistics are not included as it is intuitive from the plot that the means are the same. We have updated the figure legend which now reads

“Figure S4. No evidence for global disruption of methylation by PAE. The figure shows methylation levels averaged across CpGs in different regulatory genomic contexts. Neither

brain tissue (A & B), nor liver tissue (C & D) were grossly affected by PAE exposure (blue bars). Bars represent means and standard deviation.”